WEEK 1 · DAY 5 · OVERVIEW

Case Study

AR and ARIMA on the Curve



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ABOUT 50 MIN WITH EXERCISES / 12 SECTIONS / 5 EXERCISES

This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

Problem Statement

BUSINESS CONTEXT

Day 4 found a little memory in the daily changes and a lot in their size.

The desk now wants to know, for the curve and for credit, whether a model that remembers past changes, past surprises, or both gets closer to each date's change than Day 2's last value carried forward.

We work it through on the 10 Yr, then run the same workflow as a case study on the 10 Yr, the 2s10s slope and the synthetic IG spread.

Textbooks name these forecasting models, but here every output is a one-step value for a date inside the file and never a forecast of rates or spreads.

The held-out years stay closed.

OBJECTIVES

Fit an AR(1) by least squares and by maximum likelihood, read phi with its standard error, and take one-step values on test dates.

Set ARIMA's order and trend, choose an order with AIC and BIC over a grid, read each fit's convergence flag and roots, and check its residuals.

Walk ARIMA forward on Day 2's folds, score it with MASE on three series, and say what the models show and where they fail.

ABOUT 50 MIN WITH EXERCISES / 12 SECTIONS / 5 EXERCISES

Your Route Through Day 5

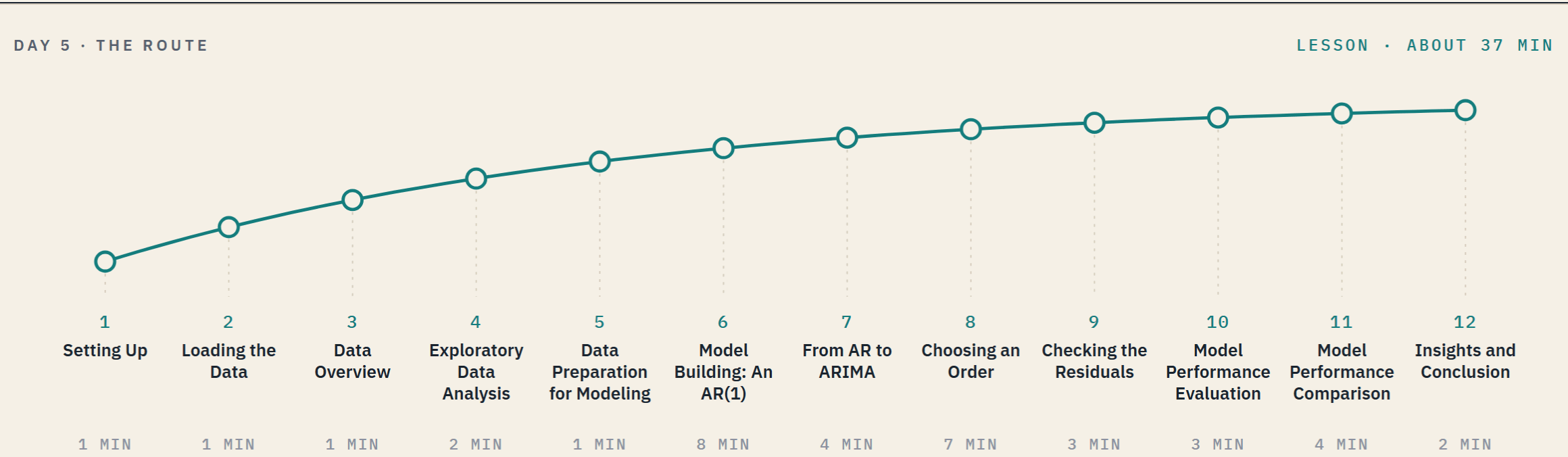


FIGURE 1 Twelve sections, in order, with about how long each takes.

Time. About 37 minutes for the lesson, then about 15 for the five exercises.

Fix the seed and load the libraries.

Setting Up

THE SETUP

```
import os
from pathlib import Path
os.environ["OMP_NUM_THREADS"] = "1"
os.environ["OPENBLAS_NUM_THREADS"] = "1"
SEED = 42
DATA = "../../../data/"
if not Path(DATA + "benchmark.csv").exists():
    DATA = "https://raw.githubusercontent.com/lenamonj/python-fixed-income/main/data/"
```

THE LIBRARIES

```
import numpy as np
import pandas as pd
from statsmodels.tsa.arima.model import ARIMA
```

Read both files and build Day 2's series.

Loading the Data

READING THE TWO FILES

```
yields = pd.read_csv(DATA + "treasury_par_yields.csv", parse_dates=["date"], index_col="date")
spreads = pd.read_csv(DATA + "course3_synthetic_spreads.csv", parse_dates=["date"], index_col="date")
CUT = "2022-12-30"
development = yields.loc[:CUT]
```

FIVE SERIES IN BP

```
levels = pd.DataFrame({"2 Yr": (development["2 Yr"] * 100).round(2),
                       "10 Yr": (development["10 Yr"] * 100).round(2),
                       "2s10s": ((development["10 Yr"] - development["2 Yr"]) * 100).round(2),
                       "IG": spreads.loc[:CUT, "ig_spread_bp"],
                       "HY": spreads.loc[:CUT, "hy_spread_bp"]})
changes = levels.diff().dropna()
print(len(levels), "levels and", len(changes), "changes, from", changes.index[0].date(), "to", changes.index[-1].date())
```

```
2001 levels and 2000 changes, from 2015-01-05 to 2022-12-30
```

Describe the three changes the case uses.

Data Overview

DESCRIBING THE THREE CHANGES

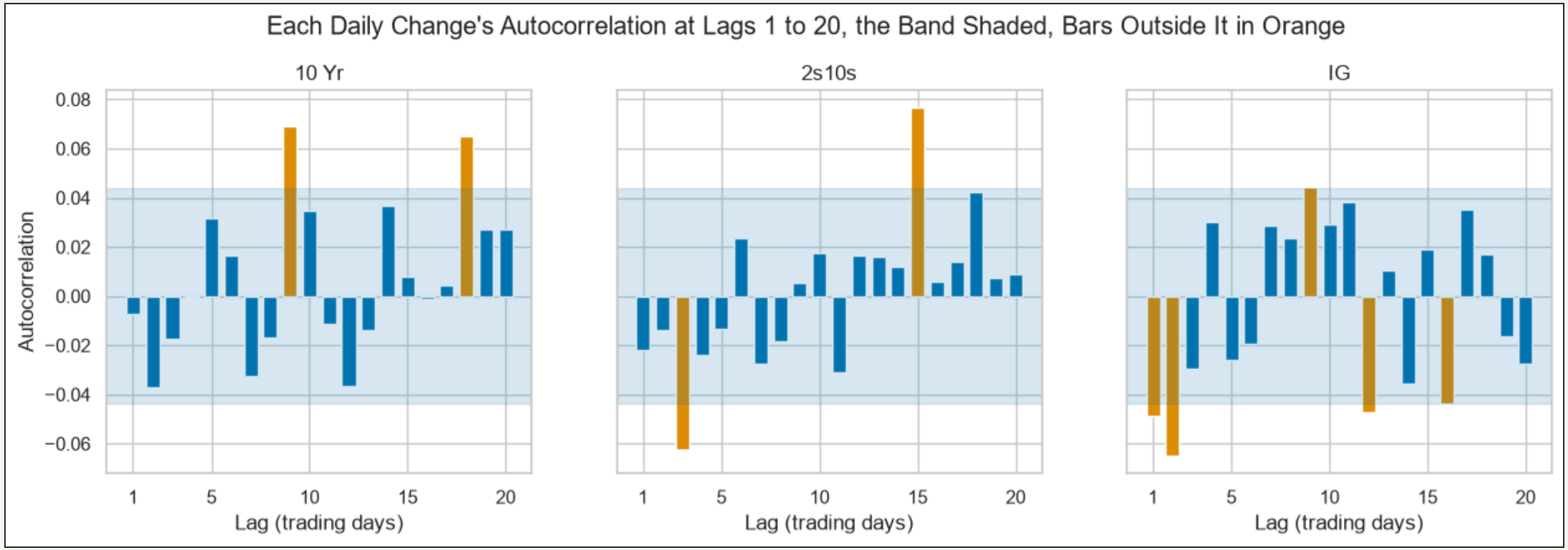
```
three = changes[["10 Yr", "2s10s", "IG"]]
overview = three.describe().T[["mean", "std", "min", "max"]]
overview["mean size"] = three.abs().mean()
overview["days with no change"] = (three == 0).sum()
overview
```

THE THREE CHANGES, IN BP

	MEAN	STD	MIN	MAX
10 Yr	0.0880	5.0285	-30.0000	29.0000
2s10s	-0.0995	3.3230	-16.0000	18.0000
IG	0.0165	1.1890	-7.0000	9.8000

Three complete series of changes, each centred near zero.

Exploratory Data Analysis



At 10 lags two of the three series hold memory a test can see. Today asks whether a model can use it.

Data Preparation for Modeling

DAY 2'S FOLDS

```
folds = list(TimeSeriesSplit(n_splits=5, test_size=252).split(changes))
last_value = pd.DataFrame({f"fold {k}": three.iloc[test_rows].abs().mean()
                           for k, (fit_rows, test_rows) in enumerate(folds, start=1)}).T
print("Average:", last_value.mean().round(3).to_dict())
print("Fold range:", (last_value.max() - last_value.min()).round(3).to_dict())
last_value
```

THE LAST VALUE'S MAE, IN BP

	10 YR	2S10S	IG
fold 1	2.6905	1.8214	0.7349
fold 2	3.4444	1.6468	0.8448
fold 3	3.6627	2.7222	0.7603
fold 4	3.3056	2.8016	1.0115
fold 5	6.2579	3.8413	1.2087

Every model below is fitted once a fold on the dates before it, and must beat these numbers by more than the folds disagree.

Model Building: An AR(1)

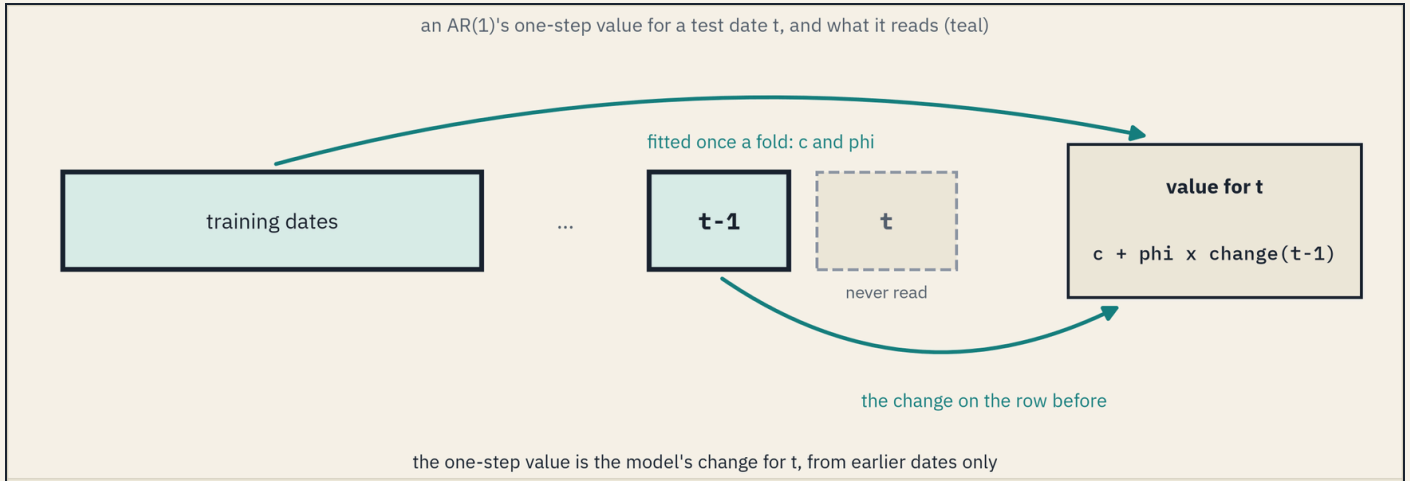


FIGURE 2 An AR(1)'s one-step value for date t: c and phi from training, times the change on t-1, never t itself.

THE SAME PHI, THREE WAYS

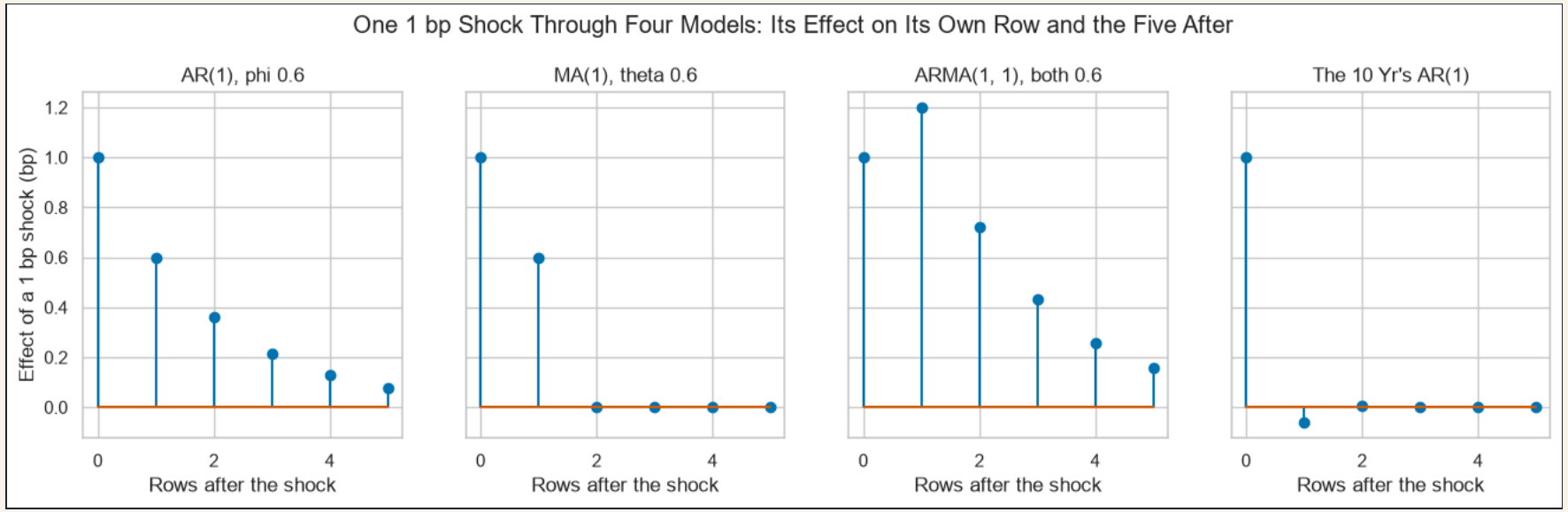
```
By hand: phi 0.620541 | intercept -0.0512
AutoReg: phi 0.620541 | intercept -0.0512
converged True | phi 0.620507 | const -0.1375
```

THE CUT TEST

```
equal: True | 1249 values compared
```

No value up to the cut moved: each one-step value reads only the row before its date.

From AR to ARIMA



FOLD 1 AS LEVELS AND AS CHANGES

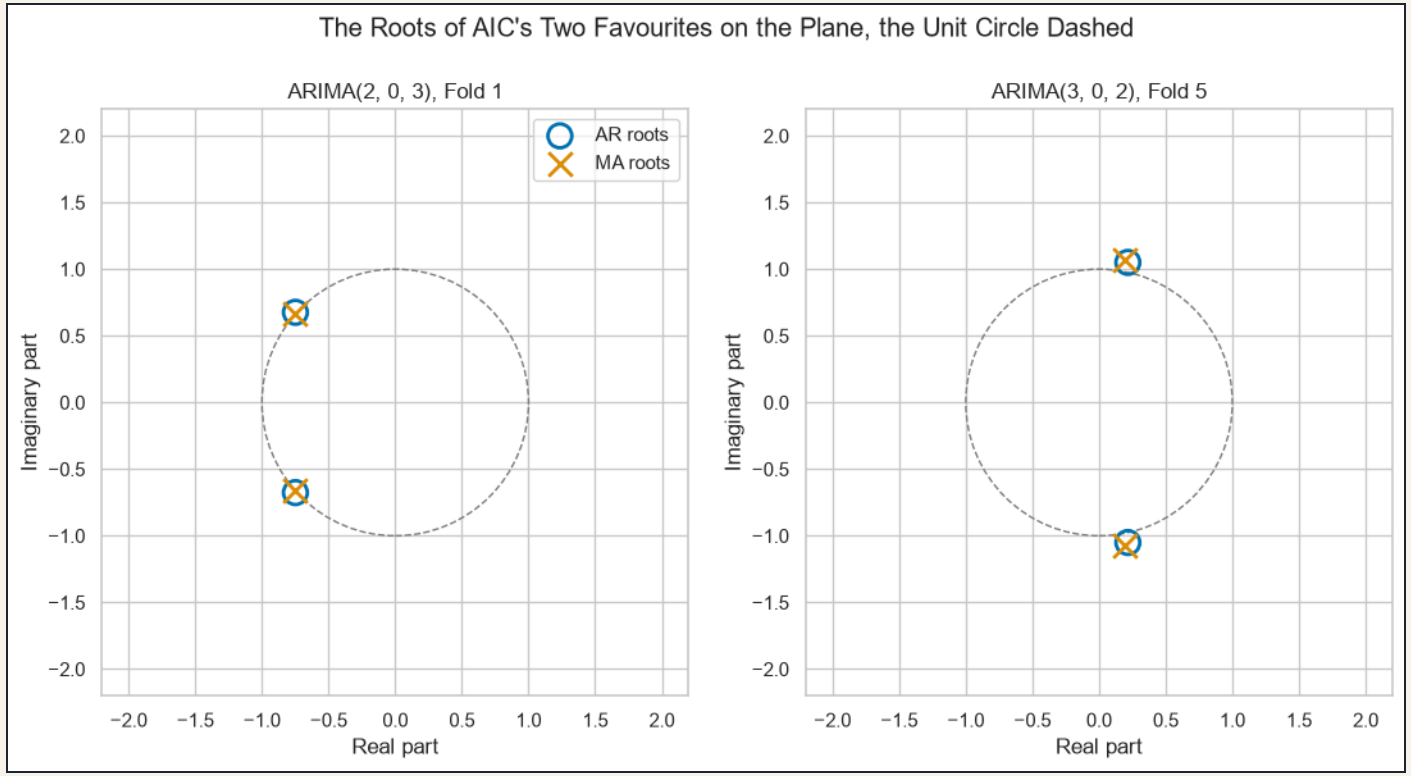
```
converged True True | phi on the levels -0.060308 | on the changes -0.060306
```

THE QUESTION

Q. Is an ARIMA(1, 1, 0) on the 10 Yr's level the same as an ARIMA(1, 0, 0) on its change?

An ARIMA with d = 1 on a level is the same model as d = 0 on its change. This course fits the changes, so every score is in bp of change.

Choosing an Order



THE QUESTION

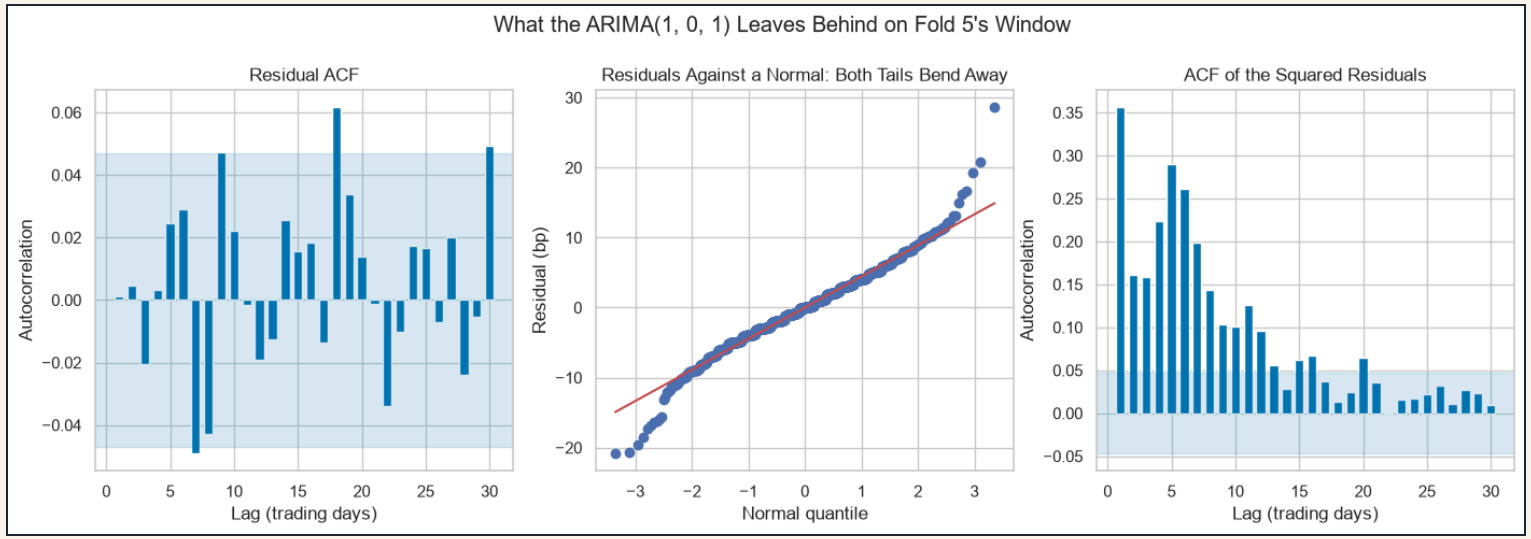
Q. Over every p and q from 0 to 4, 25 fits on fold 1's 740 training changes of the 10 Yr, which (p, q) has the lowest BIC? Write down the pair.

WHY ROOTS MATTER

A model whose AR roots all lie outside the **unit circle**, at a distance above 1, settles back after a shock.

Roots that coincide cancel: the AR part builds an echo and the MA part takes it away, so the extra terms describe nothing twice.

Checking the Residuals



WHAT THE RESIDUALS LEAVE

```
Outside the band of 0.0469 | residuals: 4 of 30 | squared: 16 of 30
Beyond 3 sd: 0.92 percent
```

LJUNG-BOX WITH TWO TERMS TAKEN OFF

```
Residuals: {10: 0.049, 20: 0.061}
The changes, the constant alone: {10: 0.046, 20: 0.06}
Squared residuals, 10 lags: 2.7e-167
```

The residuals look like the changes did: what is left is in their size, not their sign.

Model Performance Evaluation

ONE FUNCTION WALKS ANY ORDER

```
def walk_forward_arima(series, order):
    """Fit once a fold on the training dates; keep the one-step values and the MAE on the test dates."""
    maes, values, converged = [], [], []
    for fit_rows, test_rows in folds:
        train, test = changes[series].iloc[fit_rows], changes[series].iloc[test_rows]
        with warnings.catch_warnings():
            warnings.simplefilter("ignore", EstimationWarning)
            fit = ARIMA(train.to_numpy(), order=order, trend="c").fit(method_kwargs={"warn_convergence": False})
        step = fit.append(test.to_numpy(), refit=False).predict(start=len(train), end=len(train) + len(test) - 1)
        maes.append(mean_absolute_error(test, step))
        values.append(pd.Series(step, index=test.index))
        converged.append(fit.mle_retvals["converged"])
    return maes, pd.concat(values), converged
```

THE QUESTION

Q. Walked over the five folds, what is the ARIMA(1, 0, 1)'s MASE on the 10 Yr? Write down a number.

THE SCORE

MASE, Day 2's score, is a model's average fold MAE over the last value's, 3.872 bp on the 10 Yr.

Score three series, then date by date.

Model Performance Comparison

TWO MORE SERIES

```
for series in ["2s10s", "IG"]:
    for order in orders:
        maes, one_step_values[series, order], converged = walk_forward_arima(series, order)
        for k in range(5):
            rows.append({"series": series, "order": str(order), "fold": k + 1, "MAE": maes[k],
                         "last value": last_value[series].iloc[k], "converged": converged[k]})
scores = pd.DataFrame(rows)
comparison = score_table(scores)
comparison.style.format({"MAE": "{:.4f}", "MASE": "{:.4f}"}).background_gradient(subset=["MASE"], cmap="Oranges")
```

THE QUESTION

Q. On how many of the 1,260 dates is the ARIMA(1, 0, 1)'s one-step value closer to the 10 Yr's change than the last value is? Write down a number.

THE DATES SCORED

The 1,260 dates the folds score start on 2017-12-15.

Insights and Conclusion

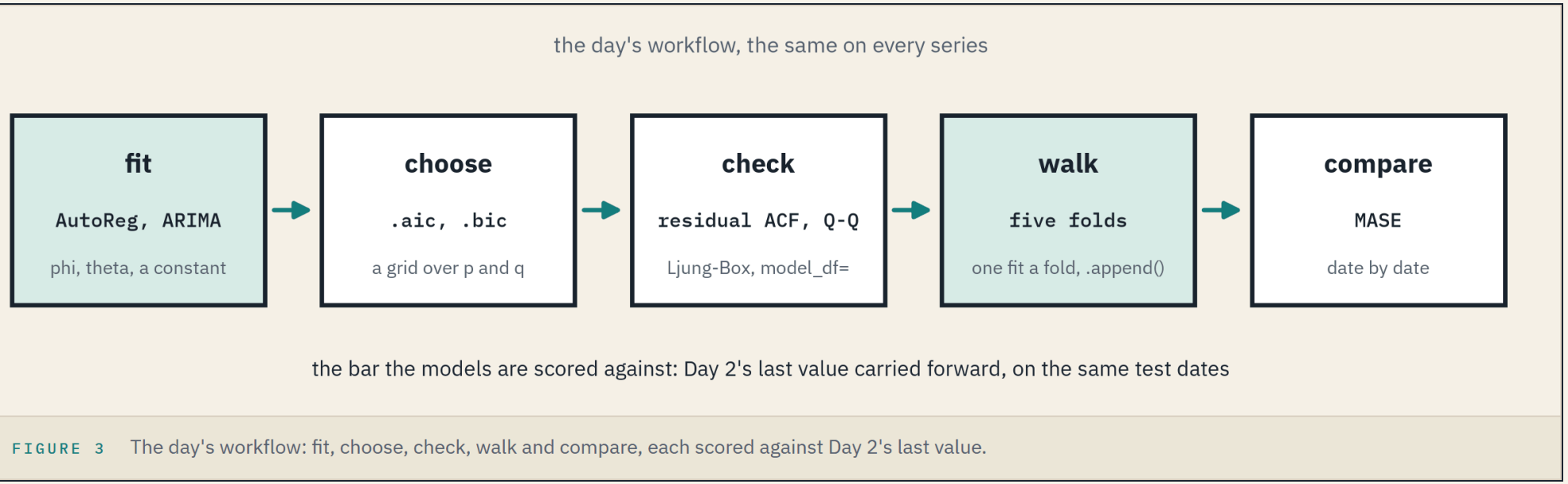


FIGURE 3 The day's workflow: fit, choose, check, walk and compare, each scored against Day 2's last value.

LEFT OPEN

What was not tested: another window, a refit on each date, and the held-out years.

You Can Now

- Fit an AR(1) by least squares and by maximum likelihood, and read phi with its standard error.
- Take one-step values from a fitted ARIMA with `.append()` and `.predict()`, and check them with the cut test.
- Set ARIMA's `order` and `trend`, and read a fit's convergence flag.
- Choose an order with AIC and BIC over a grid, and read a model's roots against the unit circle.
- Check residuals with their ACF, a Q-Q plot, and Ljung-Box with `model_df=`.
- Walk ARIMA forward on Day 2's folds, score it with MASE, and compare it with the last value date by date.

NEXT
WEEK 2 · DAY 1

Exponential Smoothing

Day 6 meets exponential smoothing, which weighs the recent past more.